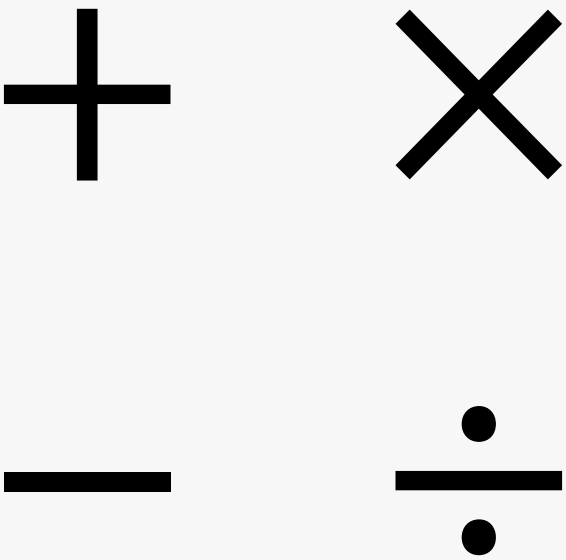


28.09.26

Lectures: Mon 11-13 MP A03
Thurs 13-15 MP A03

Problem Class: Thurs 12-13 MP A03

Office hour: Tues 15-16 B

Lecture 1

$X : \Omega \to S$

random variable X maps from sample space (Ω) to a set (S).

If $w \in \Omega$ denotes our experiment outcome then the set S can be written,
$S = \{X(w) : w \in \Omega\}$

If S is finite or countable then we may denote S as
$S = \{x_j : j = 1, \dots, J\}$

Suppose X can take uncountably many values,
then Ω is uncountable, X is cont. random variable.
$X : \Omega \to \mathbb{R}$

Distribution function

$F_X(x) = P(X \leq x), \quad -\infty < x < \infty$

$F_X : \mathbb{R} \to [0, 1]$ CDF

PDF
$f_X(x) = \frac{d F_X(x)}{dx}$

1.2 Bivariate Probability Distributions

Def 1.1
Suppose X and Y are two random variables. The joint C.D.F of X and Y is,

$$F_{X,Y}(x,y) = P(\underline{X} \leq x, \underline{Y} \leq y) \quad -\infty < x < \infty, -\infty < y < \infty$$

$$F_{X,Y}: \mathbb{R}^2 \to [0,1]$$

Def 1.2
Suppose $F_{X,Y}(x,y)$ is joint C.D.f for $\underline{X}, \underline{Y}$

If $f_{X,Y}$ exists s.t.

$$F_{X,Y}(x,y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{X,Y}(u,v)\, du\, dv$$

then $f_{X,Y}$ is joint P.D.F of X, Y

Note,

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y}$$

For joint P.D.f to be valid, following must hold,
1. $f_{X,Y}(x,y) \geq 0 \quad \forall \ (x,y) \in \mathbb{R}^2$
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty}$

Example 1.1

$$f_{X,Y}(x,y) = \begin{cases} \alpha(x+y) & 0 \le x \le y \le 1; \\ 0 & \text{otherwise} \end{cases}$$

require

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx\,dy = 1$$

$$\Rightarrow \int_0^1 \left\{ \int_0^y \alpha(x+y)\,dx \right\} dy$$

$$\Rightarrow \int_0^1 \left[\frac{\alpha x^2}{2} + \alpha x y \right]_0^y dy = 1$$

$$\Rightarrow \frac{3\alpha}{2}\int_0^1 y^2\,dy = 1$$

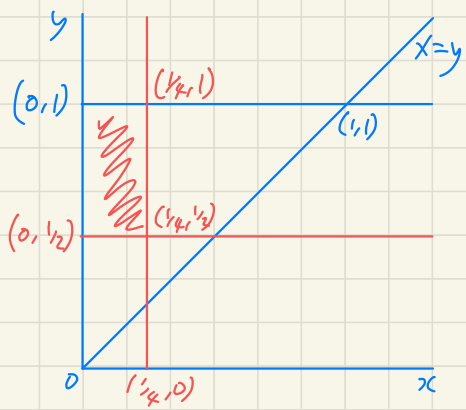
$$\Rightarrow \frac{\alpha}{2}\left[y^3\right]_0^1 = 1$$

$$\Rightarrow \frac{\alpha}{2} = 1 \Rightarrow \alpha = 2$$

b) $\mathbb{P}\left(0 \le X \le \tfrac{1}{4}, \tfrac{1}{2} \le Y \le 1\right)$

$$= \iint_C f_{X,Y}(x,y)\,dx\,dy$$

where $C = \{(x,y) \mid 0 \le x \le \tfrac{1}{4}, \tfrac{1}{2} \le y \le 1\}$



$$\int_{1/2}^{1}\int_{0}^{1/4} 2(x+y)\,dx\,dy$$

$$= \int_{1/2}^{1} 2\left[\frac{x^2}{2} + xy\right]_0^{1/4} dy = \int_{1/2}^{1} \frac{1}{16} + \frac{y}{2}\,dy$$

$$= \left[\frac{y}{16} + \frac{y^2}{4}\right]_{1/2}^{1} = \left(\frac{1}{16} + \frac{1}{4}\right) - \left(\frac{1}{32} + \frac{1}{16}\right) = \frac{7}{32}$$

Example 1.2

$$f_{X,Y}(x,y) = \begin{cases} e^{-x} & x \geq 0,\ 0 \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

$$F_{X,Y}(x,y) = \mathbb{P}(X \leq x, Y \leq y)$$

$$= \int_{-\infty}^{x} \left\{ \int_{-\infty}^{y} f_{X,Y}(u,v)\, dv \right\} du$$

$$= \int_{0}^{x} \left\{ \int_{0}^{y} e^{-u}\, dv \right\} du$$

$$= \int_{0}^{x}$$

$$F_{X,Y}(x,y) = \begin{cases} 0 & x < 0, y < 0 \\ y(1-e^{-x}) & x \geq 0 \\ & 0 \leq y \leq 1 \\ 1-e^{-x} & y > 1,\ x \geq 0 \end{cases}$$

1.2.1 Marginal p.d.f

Def 1.3:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$$

Also Note:

$$F_X(x) = F_{X,Y}(x,\infty)$$
$$F_Y(y) = F_{X,Y}(\infty,y)$$

Example 1.3

$$f_{X,Y}(x,y) = \begin{cases} e^{-x-y} & x>0,\ y>0 \\ 0 & \text{otherwise} \end{cases}$$

a) Consider $f_X(x)$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$$
$$= \int_{0}^{\infty} e^{-x-y}\,dy$$
$$= \left[-e^{-x-y}\right]_{y=0}^{\infty}$$
$$= e^{-x}$$

$$\therefore f_X(x) = \begin{cases} e^{-x} & x>0; \\ 0 & \text{otherwise} \end{cases}$$

b) Consider $f_Y(y)$ $\quad f_Y(y) = \int_{x=0}^{\infty} e^{-x-y}\,dx = e^{-y}$ $\quad f_Y(y) = \begin{cases} e^{-y} & y>0 \\ 0 & \text{otherwise} \end{cases}$

01.10.26 Workshop

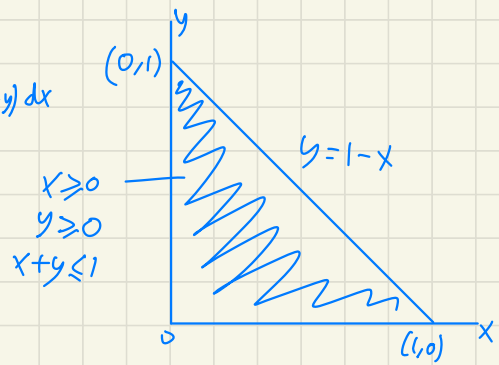
Example 1.4

$$f_{X,Y}(x,y) = \begin{cases} 24x(1-x-y) & x \geq 0,\ y \geq 0 \quad x+y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

a) Marginal Pdf of y

Integrate over x

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$$



$x+y \leq 1$
$\Rightarrow y \leq 1-x$

$$f_Y(y) = \int_0^{1-y} f_{X,Y}(x,y)\,dx$$

$$= 24\int_0^{1-y} x(1-x-y)\,dx = 24\int_0^{1-y} x - x^2 - xy\,dx$$

$$= \left[12x^2 - 8x^3 - 12x^2y\right]_0^{1-y} = 12(1-y)^2 - 8(1-y)^3 - 12y(1-y)^2$$

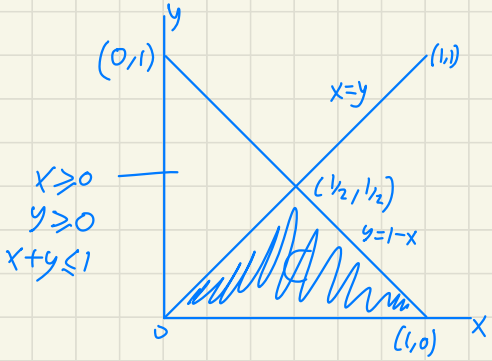
$$= 12(1-y)(1-y)^2 - 8(1-y)^3$$

$$= 4(1-y)^3$$

$$\therefore f_Y(y) = \begin{cases} 4(1-y)^3 & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

b) Calculate $\mathbb{P}(X>Y)$

Integrate $f_{X,Y}(x,y)$ over region where $X>Y$ and $X\geq 0, Y\geq 0, X+Y\leq 1$



$$\mathbb{P}(X>Y)=\iint_G f_{X,Y}(x,y)\,dx\,dy$$

$$\mathbb{P}(X>Y)=24\int_0^{1/2}\int_y^{1-y} x(1-x-y)\,dx\,dy$$

$$=24\int_0^{1/2}\left[\frac{x^2}{2}-\frac{x^3}{3}-\frac{x^2y}{2}\right]_y^{1-y}dy$$

$$=\int_0^{1/2}\left[12x^2-8x^3-12x^2y\right]_y^{1-y}dy$$

$$=\int_0^{1/2}4-12y+16y^3\,dy$$

$$=\left[4y-6y^2+4y^4\right]_0^{1/2}$$

Example 1.5

$$f_{X,Y}(x,y) = \begin{cases} 8xy & 0<y<x<1 \\ 0 & \text{otherwise} \end{cases}$$

$$f_X(x) = \int_0^x 8xy \, dy$$

$$= \left[4xy^2\right]_0^x$$

$$= 4x^3$$

$$\therefore f_X(x) = \begin{cases} 4x^3 & 0<x<1; \\ 0 & \text{otherwise} \end{cases}$$

$$f_Y(y) = \int_y^1 8xy \, dx$$

$$= \left[4x^2y\right]_y^1$$

$$= 4y - 4y^3 = 4y(1-y^2)$$

$$\therefore f_Y(y) = \begin{cases} 4y(1-y^2 & 0<y<1; \\ 0 & \text{otherwise} \end{cases}$$

If independent, $f_Y(y) \times f_X(x) = f_{X,Y}(x,y)$

$f_X(x) \cdot f_Y(y) \neq 8xy$

$\therefore$ X and Y are not independent.

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Theorem 2.1

X and Y independent random variables

$Z = X + Y$

Theorem 2.2

X and Y independent continuous random variable with P.d.f $f_X(x)$ and $f_Y(y)$

P.d.f of $Z = X + Y$ is

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z-y) f_Y(y)\, dy = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx$$

$f_Z(\,)$ is called convolution

Example 2.1

$X \sim Exp(\lambda) \Rightarrow f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$

$Y \sim Exp(\mu) \Rightarrow f_Y(y) = \begin{cases} \mu e^{-\mu y} & y > 0 \\ 0 & \text{otherwise} \end{cases}$

$\lambda > 0, \mu > 0$

X and Y are independent

Consider $Z = X + Y$

$\therefore \quad Z > 0$

Using thm 2.2 (convolution) we know that

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z-y) f_Y(y)\, dy$$

$f_X(z-y) > 0 \Leftrightarrow z - y > 0$

$\Leftrightarrow z > y$

$$f_Z(z) = \int_0^z \lambda e^{-\lambda(z-y)} \mu e^{-\mu y}\, dy$$

$$= \lambda\mu e^{-\lambda z} \int_0^z e^{-(\mu-\lambda)y}\, dy$$

$$= \lambda\mu e^{-\lambda z} \left[\frac{-1}{\mu-\lambda} e^{-(\mu-\lambda)y} \right]_0^z$$

$$= \lambda\mu e^{-\lambda z}\left[\frac{1}{\mu-\lambda}\left(1-e^{-(\mu-\lambda)z}\right)\right]$$

$$= \frac{\lambda\mu}{\mu-\lambda}\left(e^{-\lambda z}-e^{-\mu z}\right)$$

$$f_Z(z) = \begin{cases} \frac{\lambda\mu}{\mu-\lambda}\left(e^{-\lambda z}-e^{-\mu z}\right) & z>0 \\ 0 & \text{otherwise} \end{cases}$$

Thm 2.3

If X and Y ind. random variables then $g_1(X)$ and $g_2(Y)$ are independent for any "nice" functions $g_1()$ and $g_2()$.

Def 2.1

Suppose $X_1, \ldots, X_n$ is a set/collection of continuous random variables.

joint c.d.f of $X_1, \ldots, X_n$ is

$$F_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = P(X_1 < x_1, \ldots, X_n < x_n)$$

joint p.d.f of $X_1, \ldots, X_n$ is

$$f_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = \frac{\delta^n F_{X_1,\ldots,X_n}(x_1,\ldots,x_n)}{\delta x_1,\ldots,x_n}$$

$$f_X(x_1) = \int_{x_2=-\infty}^{x_2=\infty} \cdots \int_{x_n=-\infty}^{x_n=\infty} f_{X_1,\ldots,X_n}(x_1,\ldots,x_n)\,dx_2\ldots dx_n.$$

To determine

Example 2.2

$\underline{X} = (X_1, X_2, X_3)^T$

$$f_{\underline{X}}(x) = \begin{cases} 24 x_2 (1 - x_3) & 0 \le x_2 \le x_3 \le 1; \\ & 0 \le x_1 \le 1 \\ 0 & \text{otherwise.} \end{cases}$$

a) $P(X_1 < 1/2, X_2 < 1/3, X_3 < 1/2) = P$

$$P = \int_G f_{\underline{X}}(\underline{x}) \, d\underline{x}$$

$$G = \{ (x_1, x_2, x_3) \mid x_1 < 1/2, x_2 < 1/3, x_3 < 1/2 \}$$

we have $0 < x_1 < 1/2$

$x_2 < x_3 < 1$ we want $x_2 < x_3 < 1/2$

$0 < x_2 < 1/3$

So $G = \{ 0 \le x_1 \le 1/2, x_2 \le x_3 < 1/2, 0 \le x_2 < 1/3 \}$

$$P = \int_0^{1/2} \int_0^{1/3} \int_{x_2}^{1/2} f_{\underline{X}}(\underline{x}) \, dx_3 \, dx_2 \, dx_1$$

$$= \int_0^{1/2} \int_0^{1/3} \int_{x_2}^{1/2} 24 x_2 (1 - x_3) \, dx_3 \, dx_2 \, dx_1$$

$$= \int_0^{1/2} \int_0^{1/3} \left[24 x_2 x_3 - \frac{24 x_2 x_3^2}{2} \right]_{x_2}^{1/2} dx_2 dx_1$$

$$= 24 \int_0^{1/2} \int_0^{1/3} \left[x_2 x_3 - \frac{x_2 x_3^2}{2} \right]_{x_2}^{1/2} dx_2 dx_1$$

$$= 24 \int_0^{1/2} \int_0^{1/3} \frac{3}{8} x_2 - x_2^2 - \frac{x_2^4}{2} dx_2 dx_1$$

$$= 24 \int_0^{1/2} \left[\frac{3x_2^2}{16} - \frac{x_2^3}{3} - \frac{x_2^5}{10} \right]_0^{1/3} dx_1$$

$$= 24 \int_0^{1/2}$$

$$= \frac{13}{108}$$

b) $f_{X_2}(x_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{\underline{X}}(\underline{x})\, dx_3\, dx_1$

$0 \le x_1 \le 1$ limits for integration w.r.t. x_1, x_3
$x_2 \le x_3 \le 1$

$f_{X_2}(x_2) = 24 \int_0^1 \int_{x_2}^1 x_2(1-x_3)\, dx_3\, dx_1$

$= 24 \int_0^1 \left[x_2 x_3 - \frac{x_2 x_3^2}{2} \right]_{x_2}^1 dx_1$

$= 24 \int_0^1 \frac{x_2}{2} - x_2^2 + \frac{x_2^3}{2}\, dx_1$

$= 12 x_2 - 24 x_2^2 + 12 x_2^3 \left[x_1 \right]_0^1$

$= 12 x_2 (1 - 2x_2 + x_2^2)$

$= 12 x_2 (x_2 - 1)$

$$f_{X_2}(x_2) = \begin{cases} 12 x_2 (x_2 - 1) & 0 \le x_2 \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Definition 2.2
random variables $X_1, \ldots, X_n$ are
mutually independent if

$$F_{\underline{X}_{(r)}}(\underline{x}_{(r)}) = \prod_{j=1}^{|z_r|} F_{\underline{X}_{(r)j}}(x_{(r)j})$$

for each finite subset, $\underline{X}_{(r)}$ of $\{X_1, \ldots, X_2\}$

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Example 3.1

$$f_{X,Y}(x,y) = \begin{cases} x+y & 0 \le x \le 1,\ 0 \le y \le 1; \\ 0 & \text{otherwise} \end{cases}$$

a) Determine $f_{X|Y}(x|y)$

If $y \notin [0,1]$ then $f_Y(y) = 0$ and $f_{X|Y}(x|y)$ is undefined

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$f_Y(y) = \int_0^1 f_{X,Y}(x,y)\,dx = \int_0^1 x+y\,dx = \left[\frac{x^2}{2} + xy\right]_0^1$$

$$= \frac{1}{2} + y$$

$$f_{X|Y}(x|y) = \frac{x+y}{1/2+y} = \frac{2(x+y)}{1+2y}$$

For $0 \le y \le 1$:

$$f_{X|Y}(x|y) = \begin{cases} 2(x+y) & 0 \le x \le 1; \\ 0 & \text{otherwise} \end{cases}$$

b) Calc $P(X > 1/2 \mid Y = 1/3) = \int_{1/2}^1 f_{X|Y=1/3}(x|y=1/3)\,dx$

$$= \int_{1/2}^1 \frac{2(x+1/3)}{1+2/3}\,dx = \frac{6}{5}\int_{1/2}^1 x+\frac{1}{3}\,dx$$

$$= \frac{6}{5}\left[\frac{x^2}{2} + \frac{x}{3}\right]_{1/2}^{1}$$

$$= \frac{6}{5}\left(\frac{1}{2} + \frac{1}{3} - \frac{1}{8} - \frac{1}{6}\right)$$

$$= 0.65$$

Example 3.2

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{y}\exp\left[-\left(\frac{x}{y} + y\right)\right] & x>0,\ y>0. \\ 0 & \text{otherwise} \end{cases}$$

a) Calc $f_{X|Y}(x|y)$

For $y \le 0$ $f_Y(y) = 0$ $\therefore$ $f_{X|Y}(x|y)$ is not defined for $y<0$.

Consider $y>0$:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$$

$$= \int_0^{\infty} \frac{1}{y}\exp\left[-\left(\frac{x}{y} + y\right)\right]dx$$

$$= \frac{e^{-y}}{y}\int e^{-\frac{x}{y}}dx = \frac{e^{-y}}{y}\left[-y e^{-x/y}\right]_0^{\infty}$$

$$= \frac{-y e^{-y}}{y}\left[e^{-\frac{x}{y}}\right] = -e^{-y}(0-1) = e^{-y}$$

$$f_{X|Y}(x|y) = \frac{\frac{1}{y}\exp\left[-\left(\frac{x}{y}+y\right)\right]}{e^{-y}} = \frac{e^{-\frac{x}{y}}e^{-y}}{ye^{-y}} = \frac{e^{-\frac{x}{y}}}{y}$$

$\therefore$ For $y>0$,

$$f_{X|Y}(x|y) = \begin{cases} \frac{e^{-\frac{x}{y}}}{y} & x>0; \\ 0 & \text{otherwise} \end{cases}$$

b) $$P(X>1 \mid Y=y) = \int_1^\infty f_{X|Y}(x,y)\,dx = \int_1^\infty \frac{1}{y}e^{-\frac{x}{y}}\,dx$$

$$= \left[-\exp\left(\frac{-x}{y}\right)\right]_1^\infty = e^{-\frac{1}{y}}$$

Chapter 4: Expectation

$$E(X) = \int_{-\infty}^{\infty} x f_X(x)\, dx$$

If $g : \mathbb{R} \to \mathbb{R}$ is a function then

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

For $\underline{X} = (X_1, X_2, \ldots, X_n)^T$

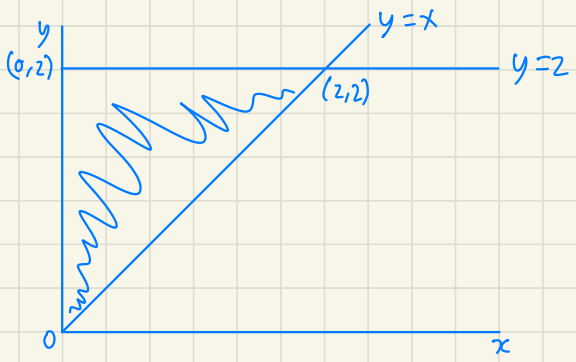
$$E(g(\underline{X})) = \int \cdots \int_{\mathbb{R}^n} \cdots g(\underline{x}) f_{\underline{X}}(\underline{x}) d\underline{x}$$

Example 4.1

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{2} xy & 0 \le x < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

Determine $E(2X+Y)$

$$E(\underbrace{2X+Y}_{g(X)}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (2x+y) f_{X,Y}(x,y)\, dx\, dy$$



$$E(2X+Y) = \int_0^2 \int_x^2 (2x+y)\,\tfrac{1}{2}\,xy\,dy\,dx$$

$$= \int_0^2 \int_x^2 x^2y + \tfrac{1}{2}xy^2\,dy\,dx = \int_0^2 \left[\frac{x^2y^2}{2} + \frac{xy^3}{6}\right]_x^2 dx$$

$$= \int_0^2 \left(2x^2 + \frac{4}{3}x - \frac{x^4}{2} - \frac{x^4}{6}\right) dx$$

$$= \left[\frac{2}{3}x^3 + \frac{4}{6}x^2 - \frac{2}{15}x^5\right]_0^2 = 3.73$$

08.10.26 Workshop

Exercise sheet 1

1. $f_X(x) = \begin{cases} \alpha x^2 & 0 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$;

$\alpha \in \mathbb{R}$

a) Det α b) Calc $P(X<1)$ c) Det $E(X)$ and $\text{Var}(X)$ d) Det CDF

a) we know $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$

$\therefore \int_0^2 \alpha x^2\,dx = 1$

$= \left[\frac{\alpha x^3}{3}\right]_0^2 = \frac{\alpha(2)^3}{3} - 0 = \frac{8\alpha}{3} = 1 \Rightarrow \alpha = \frac{3}{8}$

b) $P(X<1) = \int_0^1 \frac{3}{8}x^2\,dx = \frac{3}{8}\left[\frac{x^3}{3}\right]_0^1 = \frac{3}{8}\left[\frac{1}{3}\right] = \frac{1}{8}$

c) Determine $E[X]$ and $\text{Var}(X)$

$E(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx$

$= \int_0^2 \frac{3}{8}x^3\,dx = \frac{3}{8}\left[\frac{x^4}{4}\right]_0^2 = \frac{3}{8}\left[\frac{16}{4}\right] = \frac{12}{8} = \frac{3}{2}$

$\text{Var}(X) = E[X^2] - E[X]^2 \quad E[X^2] = \int_0^2 \frac{3}{8}x^4\,dx = \frac{3}{8}\left[\frac{x^5}{5}\right] = \frac{3}{8}\left[\frac{2^5}{5}\right] = \frac{3}{8}\left[\frac{64}{5}\right]^{8} = \frac{24}{5}$

$\text{Var}(X) = \frac{24}{5} - \left(\frac{3}{2}\right)^2 = \frac{24}{5} - \frac{9}{4} = \frac{51}{20}$

d) Determine CDF

$$\int_{-\infty}^{x} f_X(x)\,dx = \int_0^x \frac{3}{8}x^2\,dx = \frac{3}{8}\left[\frac{x^3}{3}\right]_0^x = \frac{x^3}{8}$$

2. Joint p.d.f $f_{X,Y}(x,y) = \begin{cases} 3x & 0<y<x<1; \\ 0 & \text{otherwise} \end{cases}$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$$

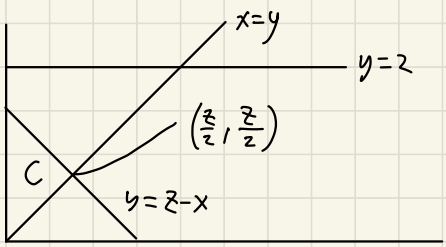
$$= \int_0^1 3x\,dy = \left[3xy\right]_0^1 = 3x$$

$$f_Y(y) = \int_0^1 3x\,dx = \left[\frac{3x^2}{2}\right]_0^1 = \frac{3}{2}$$

6. $f_{X,Y}(x,y) = \begin{cases} \frac{xy}{2} & 0<x<y<2 \\ 0 & \text{otherwise} \end{cases}$

Consider $P(Z \le z)$

$Z \le z \Rightarrow X+Y \le z \Rightarrow Y \le z-X$



$$P(Z \le z) = \iint_C f_{X,Y}(x,y)\,dy\,dx$$

08.10.26

Theorem 4.1

Proposition 4.3

If X and Y independent then
$E(XY) = E(X)E(Y)$

Def 4.2
Suppose X and Y are two random variables. Covariance of X and Y is

$$Cov(X,Y) = E[(X-E(X))(Y-E(Y))]$$

Let $E(X) = \mu_X$, $E(Y) = \mu_Y$

$$E[(X-\mu_X)(Y-\mu_Y)]$$

$$= E[XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y]$$

$$= E(XY) - \mu_X\mu_Y - \mu_Y\mu_X + \mu_X\mu_Y$$

$$= E(XY) - \mu_X\mu_Y$$

$$= E(XY) - E(X)E(Y)$$

Prop 4.4
$Cov(X,Y) = Cov(Y,X)$

Prop 4.5
If X, Y are independent, $Cov(X,Y) = 0$

Prop 4.6

For constants a, b, c, d:

$$\begin{aligned}\operatorname{Cov}(aX+b, cY+d) &= ac\left(E(XY)-E(X)E(Y)\right)\\ &= ac\operatorname{Cov}(X,Y)\end{aligned}$$

Prop 4.7

For variables $\underline{X}$ and $\underline{Y}$ with constants $\underline{a}, \underline{b}$;

$$\operatorname{Cov}\left(\left(\sum_{i=1}^{m} a_i X_i\right), \left(\sum_{j=1}^{n} b_j Y_j\right)\right) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_i b_j$$

Example 4.4

$$\operatorname{Cov}(2X+Y, 2Y-X)$$

$$\operatorname{Cov}\left(\sum_{i=1}^{n} a_i X_i, \sum_{i=1}^{m} b_i Y_j\right)$$

$$= \sum_{i=1}^{m}\sum_{j=1}^{n} a_i b_j \operatorname{Cov}(X_i, Y_j)$$

$$\operatorname{Cov}(2X+Y, 2Y-X)$$

a_1 a_2 b_1 b_2

$$(2\times 2)\operatorname{Cov}(X,Y) + (1\times 2)\operatorname{Cov}(Y,Y) - (1\times 2)\operatorname{Cov}(X,X) - (1\times 1)\operatorname{Cov}(Y,X)$$

$$= 3\operatorname{Cov}(X,Y) + 2\operatorname{Var}(Y) - 2\operatorname{Var}(X)$$

$$= 3(1) + (2\times 4) - (2\times 2) = 7$$

Lemma 4.1

If X, Y two random variables s.t. $E(X)=0$ $E(Y)=0$, $\text{var}(X)=1$ $\text{var}(Y)=1$

then, $|E(XY)| \leq 1$

Theorem 4.2

Cauchy-Bunyakovsky-Schwartz

Pair of random variables

$$[\text{cov}(X,Y)]^2 \leq \text{var}(X)\text{var}(Y)$$

Proof: Consider pair of random X, Y.

Define $\tilde{X} = \frac{X - E(X)}{\sqrt{\text{var}(X)}}$ and $\tilde{Y} = \frac{Y - E(Y)}{\sqrt{\text{var}(Y)}}$

Note $E(\tilde{X}) = 0$, $\text{var}(\tilde{X}) = 1$

$E(\tilde{Y}) = 0$, $\text{var}(\tilde{Y}) = 1$

$\tilde{X} = aX + b$ where $a = \frac{1}{\sqrt{\text{var}(X)}}$, $b = \frac{-E(X)}{\sqrt{\text{var}(X)}}$

$\tilde{Y} = cY + d$ where $c =$, $d =$

consider $\text{cov}(\tilde{X}, \tilde{Y}) = E(\tilde{X}, \tilde{Y}) - E(\tilde{X})E(\tilde{Y}) = E(\tilde{X}\tilde{Y})$

Using lemma 4.1

$|E(\tilde{X}\tilde{Y})| \leq 1$

$\Rightarrow |\text{cov}(\tilde{X}, \tilde{Y})| \leq 1$

$\Rightarrow |\text{cov}(aX+b, cY+d)| \leq 1$

$\Rightarrow |ac\ \text{cov}(X,Y)| \leq 1$ (Prop 4.6)

$$\Rightarrow \left| \frac{cov(x,y)}{\sqrt{var(x)\,var(y)}} \right| \leq 1$$

Square both sides

$$\frac{[cov(x,y)]^2}{var(y)\,var(y)}$$

Def 4.3
Pearson Correlation

$$\rho(x,y) = \frac{Cov(x,y)}{\sqrt{var(x)\,var(y)}}$$

Example 4.5

$$f_{X,Y}(x,y) = \begin{cases} x+\frac{3}{2}y^2 & 0 \le x \le 1,\ 0 \le y \le 1; \\ 0 & \text{otherwise} \end{cases}$$

Variance - covariance matrix

$$\Sigma = \begin{pmatrix} \text{Var}(x) & \text{Cov}(x,y) \\ \text{Cov}(x,y) & \text{Var}(y) \end{pmatrix}$$

$$E(xy) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f_{X,Y}(x,y)\, dy\, dx$$

$$= \int_0^1 \int_0^1 xy\left(x+\frac{3}{2}y^2\right) dy\, dx$$

$$= \int_0^1 \left[\frac{x^2y^2}{2} + \frac{3}{8}xy^4\right]_0^1 dx$$

$$= \int_0^1 \frac{x^2}{2} + \frac{3}{8}x\, dx = \left[\frac{x^3}{6} + \frac{3}{16}x^2\right]_0^1 = \frac{1}{6} + \frac{3}{16} = \frac{17}{48}$$

$$E(x) = \int_{-\infty}^{\infty} x f_X(x)\, dx$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dy = \int_0^1 x + \frac{3}{2}y^2\, dy = \left[xy + \frac{y^3}{2}\right]_0^1 = x + 1/2$$

$$\therefore f_X(x) = \begin{cases} x+1/2 & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$E(x) = \int_0^1 x^2 + \frac{x}{2}\, dx = \frac{7}{12} \qquad \text{Var}(x) = E(x^2) - E(x)^2 = \frac{5}{12} - \left(\frac{7}{12}\right)^2 = \frac{11}{144}$$

$$E(x^2) = \int_0^1 x^3 + \frac{x^2}{2}\, dx = \frac{5}{12}$$

$$E[y] = \int_{-\infty}^{\infty} y f_y(y)\, dy$$

Finish Later

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Example 5.1

$$f_{X,Y}(x,y) = \begin{cases} \frac{4x}{y} & 0 \le x \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

a) Def $E(X|Y=y) = \int_{-\infty}^{\infty} x \, f_{X|Y}(x|y)\,dx$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$$

$$f_Y(y) = \int_0^y f_{X,Y}(x,y)\,dx$$

$$= \int_0^y \frac{4x}{y}\,dx$$

$$= \left[\frac{2x^2}{y}\right]_0^y = 2y$$

$$f_Y(y) = \begin{cases} 2y & 0 \le y \le 1; \\ 0 & \text{otherwise} \end{cases}$$

$f_{X|Y}(x|y)$ def for $0 < y \le 1$

$$f_{X|Y}(x|y) = \frac{\left(\frac{4x}{y}\right)}{2y} = \frac{2x}{y^2}$$

$$f_{X|Y}(x|y) = \begin{cases} \frac{2x}{y^2} & 0 \le x \le y \\ 0 & \text{otherwise} \end{cases}$$

$$E(X|Y=y) = \int_0^y x\left(\frac{2x}{y^2}\right)dx =$$

$$\int_0^y \frac{2x^2}{y^2}\,dx = \left[\frac{2x^3}{3y^2}\right]_0^y = \frac{2y^3}{3y^2} = \frac{2y}{3}$$

$$E(X|Y=y) = \frac{2y}{3}$$

Example 5.2 (cont. 5.1)

$$f_{X,Y}(x,y) = \begin{cases} \frac{4x}{y} & 0 \le x \le y \le 1 \\ 0 \end{cases}$$

$$f_{X|Y}(x|y) = \begin{cases} \frac{2x}{y^2} & 0 \le x \le y; \\ 0 & \text{otherwise} \end{cases}$$

$$E(X|Y=y) = \frac{2}{3}y$$

$$E[X^2|Y=y] = \int_0^y x^2 \left(\frac{2x}{y^2}\right) dx = \int_0^y \frac{2x^3}{y^2} dx$$

$$= \left[\frac{2x^4}{4y^2}\right]_0^y = \frac{y^4}{2y^2} = \frac{y^2}{2}$$

For $E(X^2|Y)$ take $E(X^2|Y=y)$ replace y by Y.

$$E(X^2|Y) = \frac{Y^2}{2} \quad \text{— rand. var. } h(Y) = \frac{Y}{2}$$

$$\therefore E\left(\frac{Y^2}{2}\right) = E[E[X^2|Y]]$$

(w.r.t. $f_Y(y)$)

$$\text{Var}(X|Y=y) = E[X^2|Y=y] - [E[X|Y=y]]^2$$

Tower Property

For random X, Y;

$$E[X] = E[E[X|Y]]$$

Proof:

Consider $E(X|Y)$ ← random variable, function of Y

Let $g(y) = E(X|Y)$

Then $E[E(X|Y)] = E[g(y)]$

Since $g(y) = E(X|Y)$, consider

$$E[g(y)] = \int_{-\infty}^{\infty} g(y) f_Y(y)\, dy$$

$$= \int E(X|Y=y) f_Y(y)\, dx \quad (1)$$

$$E(X|Y=y) f_Y(y) = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\, dx \quad (2)$$

Sub (2) into (1)

$$E[g(y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\, dx$$

More Proofs

Example 5.3

$Y \sim \text{Gamma}(2, \theta)$
and $X|Y=y \sim \text{Exp}(y)$

Let $E(X)$

$$f_{X,Y=y}(x|y) = \begin{cases} ye^{-yx} & x>0; \\ 0 & \text{otherwise} \end{cases}$$

$$E[X|Y=y] = \int_0^\infty x f_{X|y}(x|y)\,dx$$

$$= \int_0^\infty xye^{-yx}\,dx$$

$$= y\int_0^\infty xe^{-yx}\,dx \quad \text{do by parts}$$

$$= \frac{1}{y}$$

Using tower property
$E[X] = E[E[X|Y]]$

$E(X|Y) = \frac{1}{Y}$

$\therefore E[X] = E\left[\frac{1}{Y}\right] = E_y\left[\frac{1}{y}\right]$

$$E\left[\frac{1}{Y}\right] = \int_0^\infty \frac{1}{y} f_y(y)\,dy$$

If $Y \sim \text{Gamma}(\alpha, \beta)$ then

$$f_y(y) = \begin{cases} \frac{\beta^\alpha y^{\alpha-1} e^{-\beta y}}{\Gamma(\alpha)} & y>0 \\ 0 & \text{otherwise} \end{cases}$$

$Y \sim \text{Gamma}(2, \theta)$

$$E(X) = \int_0^\infty \frac{1}{y}\theta^2 ye^{-\theta y}\,dy$$

$$= \theta\int_0^\infty \theta e^{-\theta y}\,dy$$

$$= \theta\left[-e^{-\theta y}\right]_0^\infty = \theta$$

Example 5.4

Sequence $X_0, X_1, X_2, \ldots$ where $X_0 = 0$

$$X_n = \begin{cases} X_{n-1} + 1 & \text{prob } 1/3 \\ X_{n-1} + 2 & \text{"} \quad \text{"} \\ X_{n-2} + 3 & \text{"} \quad \text{"} \end{cases}$$

Det $E[X_n]$

Suppose $X_{n-1} = k \quad k \in \{0, 1, 2, 3, \ldots\}$



Example 5.5

random var. X, Y s.t.

$Y \sim Exp(\lambda)$ $\quad f_Y(y) = \begin{cases} \lambda e^{-\lambda y} & y > 0; \\ 0 & \text{otherwise.} \end{cases}$

$X | Y = y \sim Poi(y)$

$P(X = x | Y = y) = \dfrac{y^x e^{-y}}{x!} \quad x \in \{0, 1, 2, \ldots\}$

Comp $P(X=0) = P(X=0 | Y=y) = e^{-y}$

Using tower property

$P(X=0) = E[P(X=0|Y)] = E[e^{-Y}]$

$= \int_0^\infty P(X=0 | Y=y) f_Y(y) \, dy$

$= \int_0^\infty e^{-(\lambda+1)}$

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Definition 5.2 (Wald's equation)

Suppose $X_1, X_2, \ldots$ sequence of i.i.d. random variables

N is non-negative integer random variable independent of X

If $T = \sum_{i=1}^{N} X_i$, then

$$E(T) = E(N)E(X_1)$$

Example 5.6

N = # eggs laid $\quad N \sim Poi(\lambda) \quad P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$

$x \in \{0, 1, 2, 3, \ldots\}$

X_i = weight of ith egg $\quad X_i \sim N(\mu, \sigma^2)$

$X_1, \ldots, X_N$ are i.i.d.

Let T = Total weight of eggs

$$= X_1 + \ldots + X_N$$

$$= \sum_{i=1}^{N} X_i$$

Since $X_1, \ldots, X_N$ are i.i.d use Wald's equation

$$E(T) = E(N)E(X_1)$$

$$= \lambda\mu$$

Definition 5.3 Conditional variance

$$\text{Var}(X|Y) = E(X^2|Y) - (E(X|Y))^2$$

Theorem 5.2

For any random variables X and Y where Var X is defined, following holds.

$$\text{Var}(X) = E(\text{Var}(X|Y)) + \text{Var}(E(X|Y))$$

Proof

$$\text{Var}(X|Y) = E(X^2|Y) - [E(X|Y)]^2 \quad (1)$$

(Function of Y)

Take E of (1) use marginal $f_Y(y)$ pdf of Y

$$E[\text{Var}(X|Y)] = E[E(X^2|Y)] - E\left[[E(X|Y)]^2\right]$$

Using tower property:

$$E(E(X^2|Y)) = E(X^2)$$

So $$E(\text{Var}(X|Y)) = E(X^2) - E\left[[E(X|Y)]^2\right] \quad (3)$$

let $g(y) = E(X|Y)$

$$\text{Var}(g(y)) = E[g(y)^2] - \{E[g(y)]^2\}$$

$$\Rightarrow E[g(y)^2] = \text{Var}(g(y)) + \{E[g(y)]^2\}$$

Since $g(y) = E(X|Y)$

$$\therefore E[(E(X|Y))^2] = \text{Var}[E(X|Y)] + \{E[E(X|Y)]\}^2$$

$$= \text{Var}[E(X|Y)] + \{E(X)\}^2 \quad \text{— Tower property}$$

$$E[\text{Var}(X|Y)] = E(X^2) - \{\text{Var}[E(X|Y)] + E(X)^2\}$$

$$= E(X^2) - \{E(X)\}^2 - \text{Var}\left[E(X|Y)\right]$$

$$\Rightarrow E\left[\text{Var}(X|Y)\right] = \text{Var}(X) - \text{Var}\left(E(X|Y)\right)$$

$$\therefore \text{Var}(X) = E\left[\text{Var}(X|Y)\right] + \text{Var}\left(E(X|Y)\right) \quad \square$$

Example 5.7

$X|Y \sim \text{Poi}(Y) \Rightarrow E(X|Y) = Y,\ \text{Var}(X|Y) = Y$

$Y \sim \text{Exp}(\lambda) \Rightarrow E(Y) = \frac{1}{\lambda},\ \text{Var}(Y) = \frac{1}{\lambda^2}$

Using Thm 5.2

$$\text{Var}(X) = E\left[\text{Var}(X|Y)\right] + \text{Var}\left[E(X|Y)\right]$$

$$= E(Y) + \text{Var}(Y)$$

$$= \frac{1}{\lambda} + \frac{1}{\lambda^2} = \frac{\lambda + 1}{\lambda^2}$$

Chapter 6

Def 6.1 Moment generating function

$M_X : A \subseteq \mathbb{R} \to [0, \infty)$

$M_X(t) = E(e^{tX})$

If $E(e^{tX})$ is not finite in open interval containing $t=0$, then MGF undefined

Example 6.1

$X \sim N(0,1) \qquad f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$

$$M_X(t) = E[e^{tX}]$$

$$= \int_{-\infty}^{\infty} e^{tx} f_X(x)\, dx$$

$$= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-x^2 + tx\right) dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\tfrac{1}{2}(x-t)^2\right) e^{\frac{t^2}{2}}\, dx$$

$$= e^{t^2/2} \int \underbrace{\frac{1}{\sqrt{2\pi}} \exp\left(-\tfrac{1}{2}(x-t)^2\right)}_{N(t,1)\ \text{p.d.f always integrates to 1.}} dx$$

$$\Rightarrow M_X(t) = e^{t^2/2} \quad t \in \mathbb{R}$$

Example 6.2

$Y \sim Exp(\lambda)$ $\quad f_Y(y) = \begin{cases} \lambda e^{-\lambda y} & y > 0; \\ 0 & \text{otherwise} \end{cases}$ $\quad \lambda > 0$ always

$M_y(t) = E(e^{ty})$

$$\int_0^\infty e^{ty} \lambda e^{-\lambda y} \, dy = \int_0^\infty \lambda e^{(ty - \lambda y)} \, dy$$

$$= \lambda \int_0^\infty e^{(t-y)y} \, dy$$

For a finite integral we require

$t - \lambda < 0$

$\Rightarrow t < \lambda$

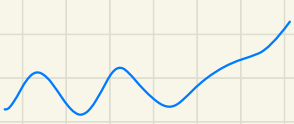
$t \in (-\infty, \lambda)$ - covers 0

$\lambda \in (t, \infty)$

$$\therefore M_y(t) = \lambda \left[\frac{1}{t-\lambda} e^{(t-\lambda)y} \right]_0^\infty$$

$$= \lambda \left(\frac{-1}{t-\lambda} \right)$$

$$= -\lambda$$



Example 6.3

$$f_X(x) = \frac{1}{\pi(1+x^2)} \qquad x \in \mathbb{R}$$

$$M_X(t) = E[e^{tX}]$$

$$= \int_{-\infty}^{\infty} \frac{e^{tx}}{\pi(1+x^2)}\, dx$$

If $t=0$, $M_X(t) = \int_{-\infty}^{\infty} \frac{1}{\pi(1+x^2)}\, dx = 1 = \frac{1}{\pi}\left[\arctan x\right]_{-\infty}^{\infty}$

If $t>0$ e^{tx}

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Prop 6.1

$M_X^{(n)}(t) = X^n e^{tX}$

If $t=0$, $M_X^{(n)}(0) = E(X^n e^0)$

$= E[X^n]$

$E(X^n)$ is n'th moment of X

$Var(X) = E(X^2) - (E(X))^2$

$E(X^2) = M_X^2(0)$ } diff w.r.t. t

$E(X) = M_X^1(0)$ }

$\therefore Var(X) = M_X^2(0) - \{M_X^1(0)\}^2$

Example 6.2 revisited

$M_Y(t) = \frac{\lambda}{\lambda - t}$

$M_Y'(t) = \lambda(\lambda - t)^{-2}$

$M_Y''(t) = 2\lambda(\lambda - t)^{-3}$

$\therefore M_Y'(0) = \frac{1}{\lambda}$, $M_Y''(0) = \frac{2}{\lambda^2}$

$Var(Y) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$

Example 6.4.

$Y \sim Exp(\lambda)$

$$M_y(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda$$

Consider derivatives of $M_y(t)$

Let $M_y^{(n)}(t)$ be n'th derivative w.r.t. t.

$$M_y^{(1)}(t) = \lambda(\lambda - t)^{-2}$$
$$M_y^{(2)}(t) = 2\lambda(\lambda - t)^{-3}$$
$$M_y^{(3)}(t) = 3!\lambda(\lambda - t)^{-4}$$

Generally,
$$M_y^{(n)}(t) = n!\lambda(\lambda - t)^{-(\lambda+1)}$$

Taylor series about 0 is

$$M_y(t) = \sum_{n=0}^{\infty} \frac{M_y^{(n)}(0)}{n!} t^n$$

$$= 1 + \frac{t}{\lambda} + \frac{t^2}{\lambda^2} + \frac{t^3}{\lambda^3} + \cdots$$

$$= \sum_{n=0}^{\infty} \left(\frac{t}{\lambda}\right)^n$$

Thm 6.2

random variable X has MGF $M_X(t)$, for constants $a, b \in \mathbb{R}$, the MGF of $Y = a + bX$ is,

$$M_Y(t) = e^{at} M_X(bt)$$

Def 6.2

Suppose $\underline{X} = (X_1, \ldots, X_n)^T$ multivariate r.v. with joint p.d.f f_X

Example 6.5

Suppose $X_i \sim N(\mu, \sigma_i^2)$ and $X_1, \ldots, X_n$ mutually independent

Let $X \sim N(\mu, \sigma^2)$

Consider $E[e^{tX}] = M_X(t)$

$$M_X(t) = E[e^{tX}]$$

$$= \int_{-\infty}^{\infty} e^{tx} f_X(x)\,dx$$

$$= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{\frac{-1}{2\sigma^2}(x-\mu)^2\right\} dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{\frac{-1}{2\sigma^2}(x-\mu)^2 + tx\right\} dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{\frac{-1}{2\sigma^2}(x^2 - 2\mu x + \mu^2) + tx\right\} dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{\frac{-1}{2\sigma^2}(x^2 - 2(\mu+\sigma^2 t)x + \mu^2\right\} dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{\frac{-1}{2\sigma^2}(x-(\mu+\sigma^2 t))^2\right\} dx \cdot \exp\left\{\frac{-1}{2\sigma^2}(-2\mu\sigma^2 t - \sigma^4 t^2)\right\}$$

$$= \exp\left\{\mu t + \frac{1}{2}\sigma^2 t^2\right\} \times \int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{\frac{-1}{2\sigma^2}(x-(\mu+\sigma^2 t))^2\right\} dx}_{\text{Pdf of a } N(\mu+\sigma^2 t, \sigma^2) \text{ distribution} \therefore \text{integrates to 1}}$$

$$= \exp\left\{\mu t + \frac{1}{2}\sigma^2 t^2\right\}$$

$\therefore$ If $X \sim N(\mu, \sigma^2)$

then $M_X(t) = \exp\left\{\mu t + \frac{1}{2}\sigma^2 t\right\}$

$X_i \sim N(\mu_i, \sigma_i^2)$ then $M_{X_i}(t) = \exp\left\{\mu_i t + \frac{1}{2}\sigma_i^2 t\right\}$

Consider $Y = \sum_{i=1}^{n} c_i X_i$

$$M_Y(t) = \mathbb{E}[e^{tY}]$$

$$= \mathbb{E}\left[e^{t\sum_{i=1}^{n} c_i X_i}\right]$$

$$= \mathbb{E}[e^{t(c_1 X_1, \ldots c_n X_n)}]$$

$$= \mathbb{E}[e^{tc_1 X_1} e^{tc_2 X_2} \times \cdots \times e^{tc_n X_n}]$$

$$= \prod_{i=1}^{n} \mathbb{E}[e^{tc_i X_i}] \quad \text{(indep. as } X_1, \ldots, X_n\text{)}.$$

$$= \prod_{i=1}^{n} M_{X_i}(c_i t)$$

$$= \prod_{i=1}^{n} \exp\left\{\mu_i c_i t + \frac{1}{2}\sigma_i^2 c_i^2 t^2\right\}$$

$$= \exp\left\{\sum_{i=1}^{n} \mu_i c_i t + \sum_{i=1}^{n} \frac{1}{2}\sigma_i^2 c_i^2 t_i^2\right\}$$



Example 6.6

$$M_X(t) = E[e^{tX}]$$

$$= \int_{-\infty}^{\infty} e^{tx} f_X(x)\,dx$$

$$= \int_0^{\infty} e^{tx} \frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2}\,dx$$

$$= \int_0^{\infty} \frac{1}{2^{\nu/2}\Gamma\left(\frac{\nu}{2}\right)} x^{\nu/2-1} e^{(-1/2)(1-2t)x}\,dx$$

If $1-2t \le 0$ then integral not finite

$1-2t > 0$ " " is finite

$\Rightarrow t < 1/2$

Let $u = \frac{1}{2}(1-2t)x$

$\Rightarrow x = \frac{2u}{1-2t}$, $dx = \frac{2}{1-2t}\,du$

$$\therefore M_X(t) = \int_0^{\infty} \frac{1}{2^{\nu/2}\Gamma(\nu/2)} \left(\frac{2u}{1-2t}\right)^{\nu/2}$$

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Exercise sheet 3

1. $X \sim N(0,1)$ and $Y|X \sim N(0,X)$

a) $f_X(x) = \begin{cases} 1 & 0<x<1; \\ 0 & \text{otherwise} \end{cases}$

$f_{Y|X}(y|x) = \begin{cases} \frac{1}{x} & 0<y<x; \\ 0 & \text{otherwise} \end{cases}$

b) $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_X(x)}$

we have $f_{Y|X}(y|x)$ and $f_X(x)$,

$f_{Y|X}(y|x) f_X(x) = f_{X,Y}(x,y)$

$f_{X,Y}(x,y) = \frac{1}{x} \cdot 1 = \frac{1}{x}$

$f_{X,Y}(x,y) = \begin{cases} \frac{1}{x} & 0<y<x<1; \\ 0 & \text{otherwise} \end{cases}$

$f_Y(y) = \int_y^1 \frac{1}{x}\,dx = [\ln x]_y^1 = \ln 1 - \ln y = -\ln(y)$

$f_Y(y) = \begin{cases} -\ln(y) & 0<y<1; \\ 0 & \text{otherwise} \end{cases}$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$= \frac{\frac{1}{x}}{-\ln(y)} = \frac{-1}{x\ln(y)}$$

$$f_{X|Y}(x|y) = \begin{cases} \frac{-1}{x\ln(y)} & y < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

c) Tower Property:

$E(X) = E(E(X|Y))$

$E(Y) = E(E(Y|X))$

$$E(Y|X) = \int_0^X y\left(\frac{1}{x}\right)dy = \left[\frac{y^2}{2x}\right]_0^X = \frac{x^2}{2x} = \frac{x}{2}$$

$$E\left(\frac{X}{2}\right) = \int_0^1 \left(\frac{x}{2}\right)(1)\,dx = \left[\frac{x^2}{4}\right]_0^1 = \frac{1}{4}$$

$\therefore E(Y) = \frac{1}{4}$

2.

X	£50	£100	£150
$P(X_i = x)$	$1/2$	$1/3$	$1/6$

3. $X \sim Exp(1)$
$Y|X \sim Poi(X)$

a) Determine $P(Y \leq 2)$

$$P(Y \leq 2) = \int_0^\infty P(Y \leq 2 | X = x) f_X(x)\, dx$$

$$= \int_0^\infty \left[P(Y=0|X=x) + P(Y=1|X=x) + P(Y=2|X=x) \right] f_X(x)\, dx$$

$Y|X=x \sim Poi(x)$

$$P(Y=y|X=x) = \frac{x^y e^{-x}}{y!}$$

$$f_X(x) = \begin{cases} e^{-x} & x > 0; \\ 0 & \text{otherwise} \end{cases}$$

$$P(Y \leq 2) = \int_0^\infty \left(e^{-x} + xe^{-x} + \frac{x^2}{2} e^{-x}\right) e^{-x}\, dx$$

22.10.26

Example 6.7

$X_i \sim \chi^2_{v_i}$ $X_1, \dots, X_n$ mutually independent

$$S = \sum_{i=1}^{n} X_i$$

From ex 6.6 $M_{X_i}(t) = \frac{1}{(1-2t)^{v_i/2}}$ $t < 1/2$

Consider

$$M_S(t) = E[e^{tS}]$$
$$= E[e^{t(X_1 + \dots + X_n)}]$$
$$= E[e^{tX_1} e^{tX_2} \dots e^{tX_n}]$$
$$= \prod_{i=1}^{n} E[e^{tX_i}] \quad \text{(because } X_i \text{ independent)}$$
$$= \prod M_{X_i}(t)$$
$$= \prod_{i=1}^{n}$$

Definition 6.3 (Probability generating function)
r.v. X takes values in subset $\{0,1,2,\ldots\}$, p.m.f. $p_X(x)$

The p.g.f. of X, $\varphi_X : [0,1] \to [0,1]$ is defined

$$\varphi_X(s) = E[s^X] = \sum_{i=0}^{\infty} s^i p_X(x)$$

Prop 6.2
$\varphi_X(s) = M_X(\log s)$

$M_X(t) = \varphi_X(e^t)$

Example 6.8

Show $E(X) = \varphi_X'(1)$

$$\varphi_X(s) = \sum_{x=0}^{\infty} s^x P(X=x)$$

Take derivative wrt s

$$\varphi_X'(s) = \frac{d}{ds} \sum_{x=0}^{n} s^x P(X=x)$$

$$= \sum_{x=0}^{n} \frac{d}{ds}(s^x) P(X=x)$$

$$= \sum_{x=0}^{n} x s^{x-1} P(X=x)$$

If $s=1$, then

$$\varphi_X'(1) = \frac{\delta}{ds} \varphi(x) \Big|_{s=1}$$

Chapter 7 inequalities

Markov's inequality

Suppose $X \geq 0$ is a r.v. and $C > 0$ is a constant

$$P(X \geq C) \leq \frac{E(X)}{C}$$

Proof:

let Y be r.v. s.t

$$Y = \begin{cases} 1 & \text{if } X \geq C \\ 0 & \text{if } X < C \end{cases}$$

let $Y \sim \text{Bernoulli}(p)$, where $p = P(X \geq C)$ ($p = P(Y=1)$)

Consider $E(Y) = \sum_y y P(Y=y)$

$$= 0 \times P(Y=0) + 1 \times P(Y=1) = P(Y=1) = P(X \geq C)$$

Consider $X \geq C$,

If $X \geq C \Rightarrow X \geq CY$

$\Rightarrow E(X) \geq E(CY)$

$\Rightarrow E(X) \geq C E(Y)$

$\therefore E(Y) \leq \frac{E(X)}{C}$

$\Rightarrow P(X \geq C) \leq \frac{E(X)}{C}$

Example 7.1

Let X = Annual income of randomly selected worker.

Chebyshev's inequality

If X is a random variable with mean μ, and variance

Proof

Example 7.2

$X \sim Exp(\frac{1}{2})$

Seek an upper bound for $P(X \geq 10)$

i) Markov's inequality

$X \geq 0$, $C > 0$

$P(X \geq C) \leq \frac{E(X)}{C}$

Since $X \sim Exp(\frac{1}{2})$

$E(X) = 2$ $\quad \left(E(X) = \int_0^\infty \frac{x}{2} e^{-\frac{1}{2}x} dx \right)$

Choose $C = 10$,

$P(X \geq 10) \leq \frac{2}{10} = \frac{1}{5}$

ii) Chebyshev's inequality

$P(|X - \mu| \geq C) \leq \frac{\sigma^2}{C^2}$

If $X \sim Exp(\frac{1}{2})$, $Var(X) = 4 = \sigma^2$

Here $\mu = 2$

$\therefore P(|X - 2| \geq C) \leq \frac{4}{C^2}$

$P\left(\{X - 2 \leq -C\} \cup \{X - 2 \geq C\} \right) \leq \frac{4}{C^2}$

$$\Rightarrow P\left(\{X \leq -c+2\} \cup \{X \geq c+2\}\right) \leq \frac{4}{c^2}$$

Choose $c=8$:

$$\Rightarrow P\left(\{X \leq -8+2\} \cup \{X \geq 8+2\}\right) \leq \frac{4}{64} = \frac{1}{16}$$

$$\Rightarrow P\left(\{X \leq -6\} \cup \{X \geq 10\}\right) \leq \frac{1}{16}$$

$$\Rightarrow P(X \leq -6) + P(X \geq 10) \leq \frac{1}{16}$$

$P(X \leq -6)$: 0, exp doesn't take -ve values

$\therefore P(X \leq -6) = 0$

$$\Rightarrow P(X \geq 10) \leq \frac{1}{16}$$

Chebyshev better.

Definition 7.3
Weak law of large numbers

Suppose $X_1, \ldots, X_n$ i.i.d. r.v
$E(X_i) = \mu$, $\text{Var}(X_i) = \sigma^2 < \infty$ $(i \in \{1, \ldots, n\})$. Then, for any constant $\epsilon > 0$.

$$\lim_{n \to \infty} P\left(\left| \frac{X_1 + \cdots + X_n}{n} - \mu \right| > \epsilon \right) = 0$$

Theorem 7.1
Strong law of large numbers

Suppose $X_1, \ldots, X_n$ i.i.d. r.v
$E(X_i) = \mu$, $\text{Var}(X_i) = \sigma^2 < \infty$ $(i \in \{1, \ldots, n\})$. Then

$$P\left(\lim_{n \to \infty} \frac{X_1 + \cdots + X_n}{n} = \mu \right) = 1$$

Def 7.4 (Convergence in Distribution)

Thm 7.2

Example 7.3

Suppose that $X_n \sim Bin\left(n, \frac{\lambda}{n}\right)$ $(0 < \lambda < n)$
also $X \sim Pa(\lambda)$

Show $X_n \xrightarrow{p} X$

Suppose $y \sim Bin(n, p)$

$M_y(t) = E[e^{ty}]$

$$= \sum_{y=0}^{\infty} e^{ty} P(Y=y)$$

$$= \sum_{y=0}^{n} e^{ty} \binom{n}{y} p^y (1-p)^{n-y}$$

$$= \sum_{y=0}^{n} \binom{n}{y} p e^t$$

CLT

Example 7.4

r.v $X_1, \ldots, X_n$ i.i.d. $\quad E(X_i) = 2$

$X_i \sim Exp(\frac{1}{2}) \quad Var(X_i) = 4$

$i = 1, \ldots, n$

Desire $Y = \sum_{i=1}^{50} X_i$

Use C.L.T, Since n is large.

$$\frac{Y - n\mu}{\sigma\sqrt{n}} \sim N(0,1)$$

Using $n = 50, \mu = 2 \quad \sigma = 2$

$$\frac{Y - (50 \times 2)}{2\sqrt{50}} \sim N(0,1)$$

$$P(90 < Y < 110) = P\left(\frac{90-100}{2\sqrt{50}} < \frac{Y-100}{2\sqrt{50}} < \frac{110-100}{2\sqrt{50}}\right)$$

$$= P\left(\frac{-\sqrt{2}}{2} < Z < \frac{\sqrt{2}}{2}\right)$$

Where $Z \sim N(0,1)$

$$= 2\Phi\left(\frac{\sqrt{2}}{2}\right) - 1 = 0.520$$